

(Time: 2 hours)

[Total Marks: 60]

- N. B.: (1) **All** questions are **compulsory**.
 (2) Make **suitable assumptions** wherever necessary and **state the assumptions** made.
 (3) Answers to the **same question** must be **written together**.
 (4) Numbers to the **right** indicate **marks**.
 (5) Draw **neat labeled diagrams** wherever **necessary**.
 (6) Use of **Non-programmable** calculator is **allowed**.

1. **Attempt any two of the following:** 12
 - a. With the help of a neat diagram, explain the logical design of IoT.
 - b. Explain the push-pull communication model.
 - c. Discuss the wearable devices as application of IoT.
 - d. How is IoT applied in smart cities? Explain.
 2. **Attempt any two of the following:** 12
 - a. Explain the fog based architecture of IoT. What are the advantages of fog computing?
 - b. What is Social Internet of Things? What are its components?
 - c. What is Raspberry Pi? What are the hardware requirements for Raspberry Pi? Why is it used?
 - d. Explain the Arduino programming with example.
 3. **Attempt any two of the following:** 12
 - a. Explain full virtualization and para virtualization.
 - b. What is Hardware assisted virtualization? Explain.
 - c. What are different trust properties? Discuss.
 - d. What are the different privacy challenges? Explain with example.
 4. **Attempt any two of the following:** 12
 - a. How IoT can be used for (i) Remote operations in construction (ii) Logistics supply in military (iii) Remote operations in construction?
 - b. How can IoT help in agriculture? Explain.
 - c. Distinguish between cloud and IoT.
 - d. Define and explain the term Cloud Computing as defined by NIST.
 5. **Attempt any two of the following:** 12
 - a. What is a smart city? How can the smartness of smart cities be measured?
 - b. Discuss the adoption of IoT in automation.
 - c. Explain how IoT can be used in the Avoid-Shift-Improve (ASI) framework proposed by the International Energy Agency (IEA) to improve transportation in a country.
 - d. Describe a futuristic situation where you will use IoT to provide solution to a problem (either an old or emerging problem) within your community or country.
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